

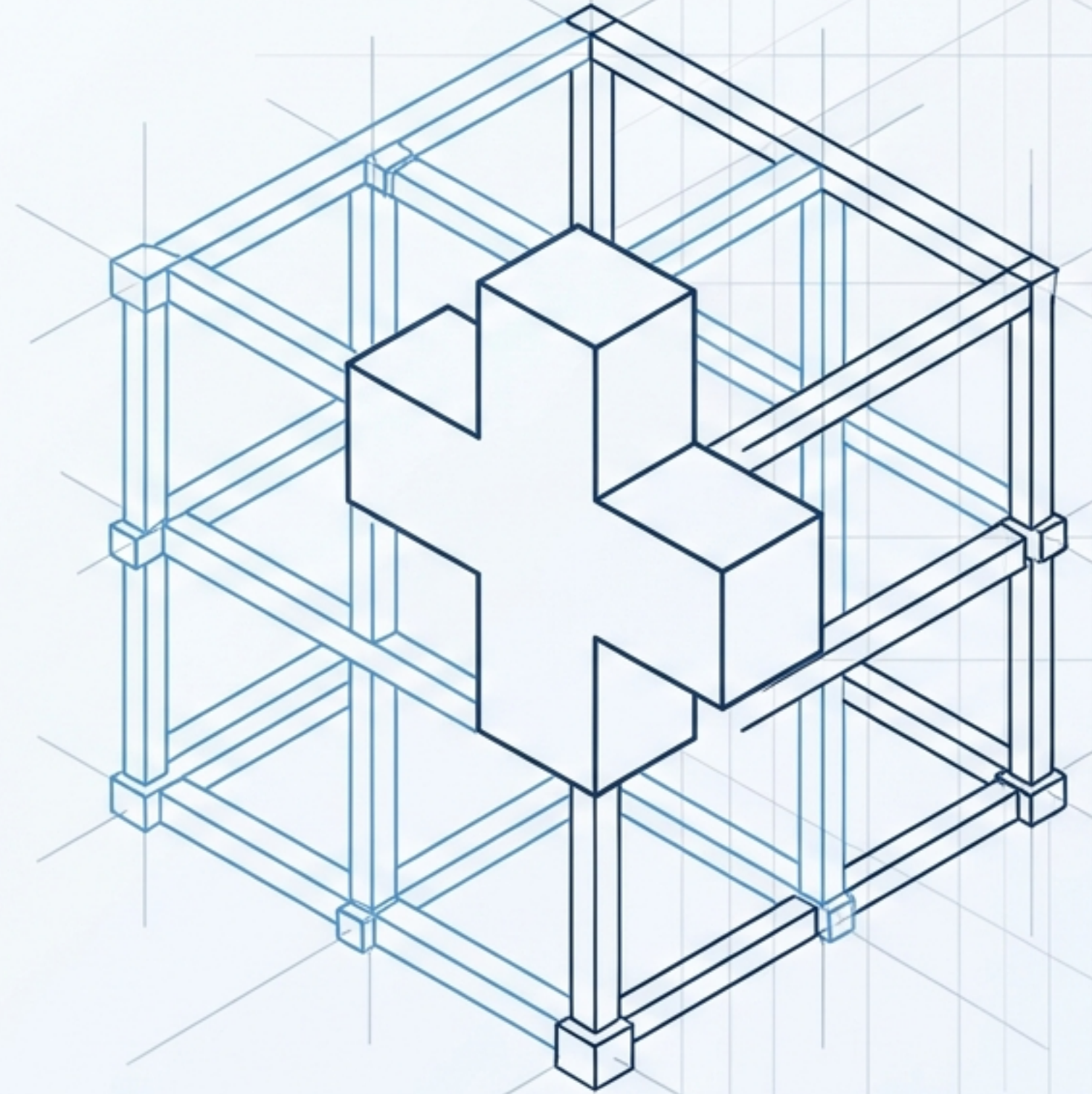
Fracto - Doctor Appointment Booking System

Submitted by: Hasvvath.R

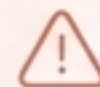
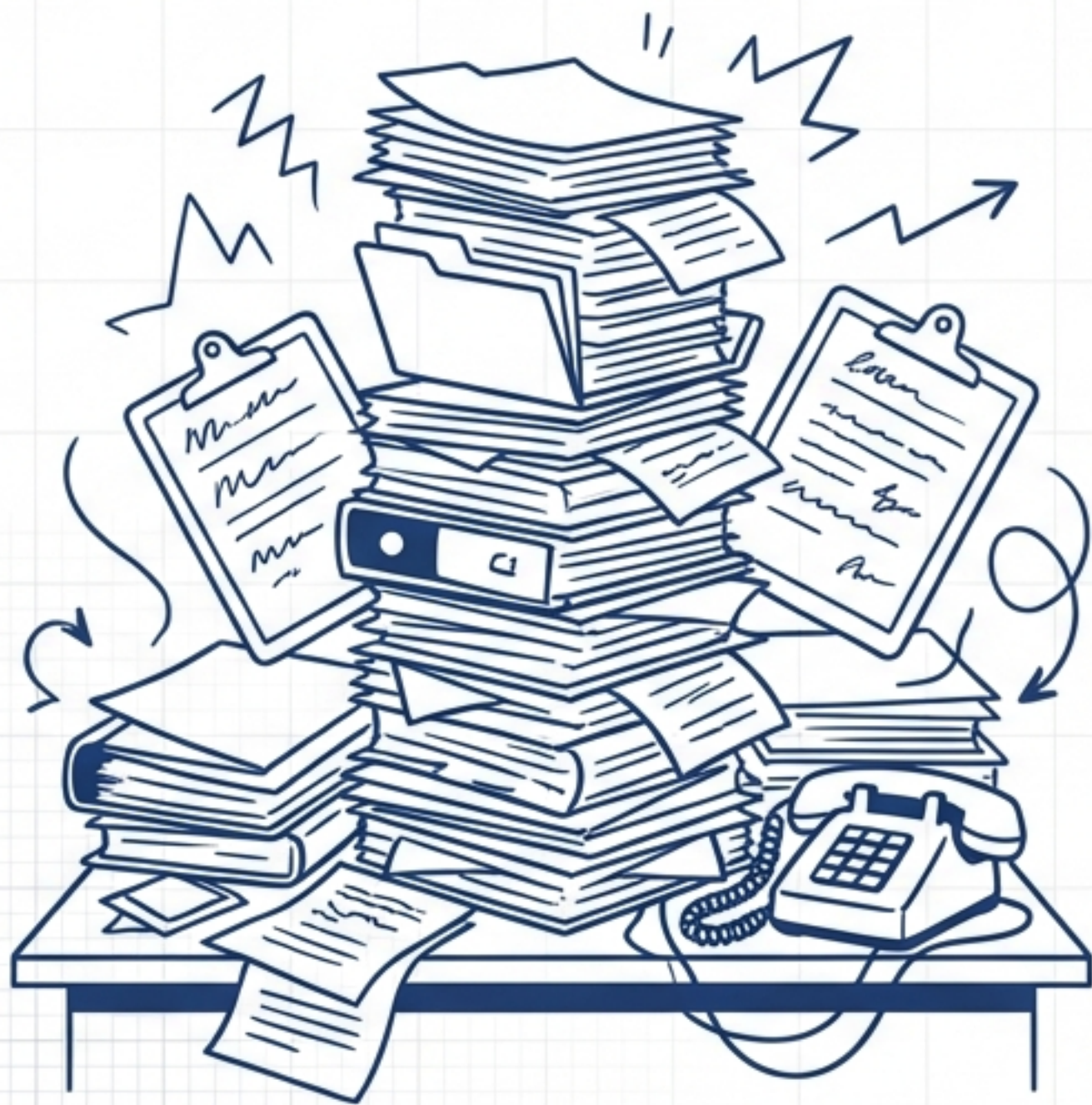
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Batch Name: WIPRO NGA - .Net Full Stack Angular - FY26

Instructor Name: Mr. Ramesh Nediyaath

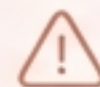


Understanding the Challenges in Healthcare Booking



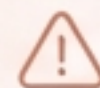
Manual Inefficiency

Traditional systems cause long waits, duplicate bookings, and disorganized clinic schedules.



Patient Friction

Patients face extreme difficulty finding available doctors, tracking down the right specializations, and securing convenient time slots.



Management Struggles

Hospitals and clinics lack centralized, digital tools to efficiently track patients, update statuses, and oversee daily appointments.

Introducing the Digital Appointment Platform

■ Meet Fracto:

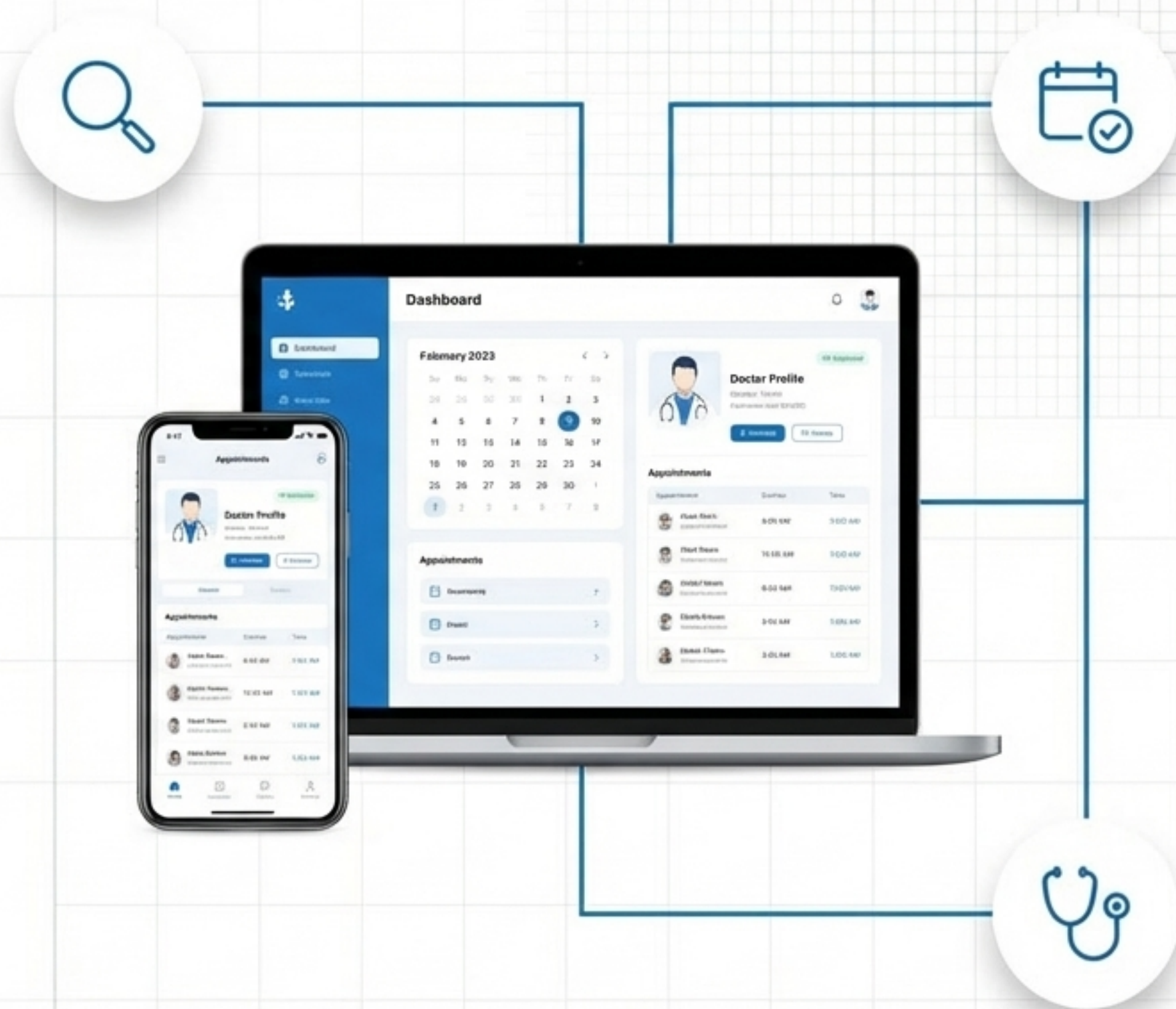
A comprehensive, full-stack online platform designed to bridge the gap between patients and healthcare providers.

■ Core Mission:

Digitize and simplify the entire lifecycle of a medical appointment—from the initial search to the final booking and management.

■ Value Proposition:

Delivers real-time availability, highly secure data handling, and an effortless administrative experience.



What the System Aims to Achieve



User-Friendly Booking

Enable patients to easily search and filter doctors by city, specialization, and user ratings.



Structured Scheduling

Enforce strict slot capacities, manage real-time availability, and entirely prevent duplicate bookings.



Secure Access

Protect sensitive patient and doctor data using robust JWT (JSON Web Token) authentication.



Scalable Tech

Build a resilient, full-stack digital infrastructure using modern, industry-standard frameworks.

The Building Blocks of the Application



Frontend (The User Interface)

- ✓ Built with Angular and TypeScript.
- ✓ Uses NgRx for centralized state management.
- ✓ Relies on Axios for seamless API communication.



Backend (The Core Logic)

- ✓ Powered by ASP.NET Core Web API.
- ✓ Secured via JWT Authentication.
- ✓ Tested and documented using Swagger.



Database (The Storage Center)

- ✓ Housed in Microsoft SQL Server.
- ✓ Managed via Entity Framework Core (ORM) for efficient data mapping.

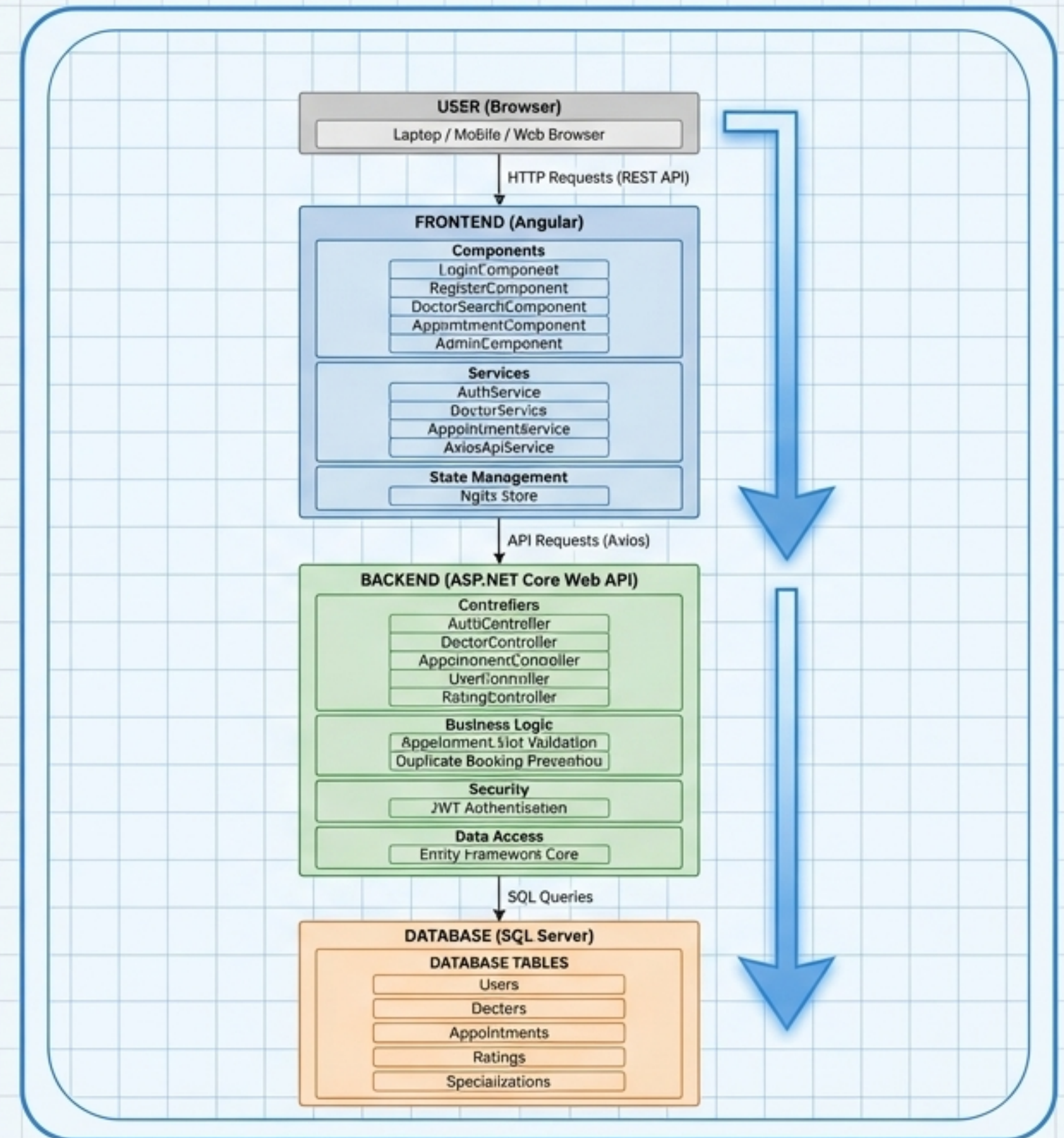
The Three-Tier Application Structure

A Scalable Design

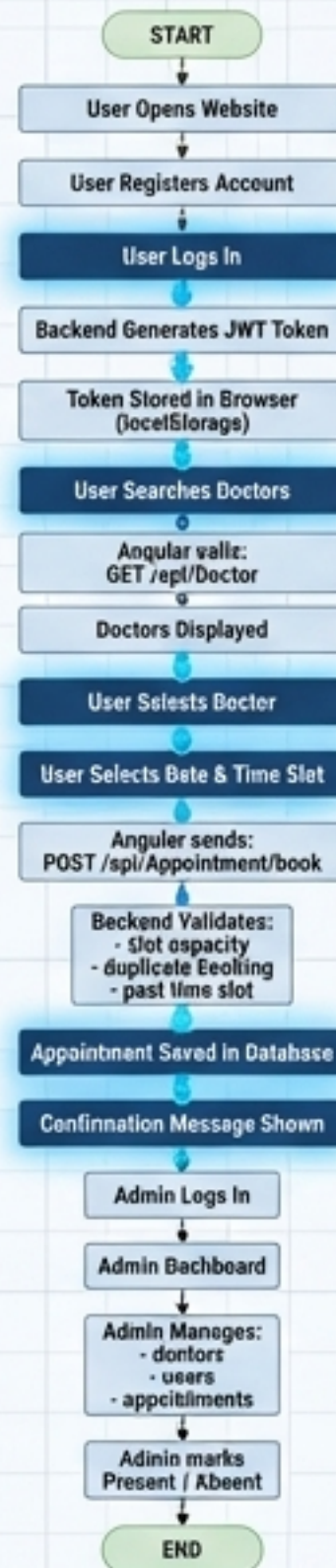
The system separates concerns into three distinct layers: the Angular Frontend (UI), the ASP.NET Backend (Logic), and the SQL Server Database (Storage).

The Flow of Data

User actions trigger API requests via Axios, which are processed by backend controllers, validated by business logic, and finally queried in the database via Entity Framework Core.



The User Journey from Login to Booking



Step-by-Step Logic:

1

Authenticate: User logs in and receives a secure JWT token.

2

Discover: Angular calls the GET /api/Doctor endpoint to display available professionals.

3

Select & Validate: User selects a time slot. The backend strictly validates capacity and prevents duplicate or past bookings.

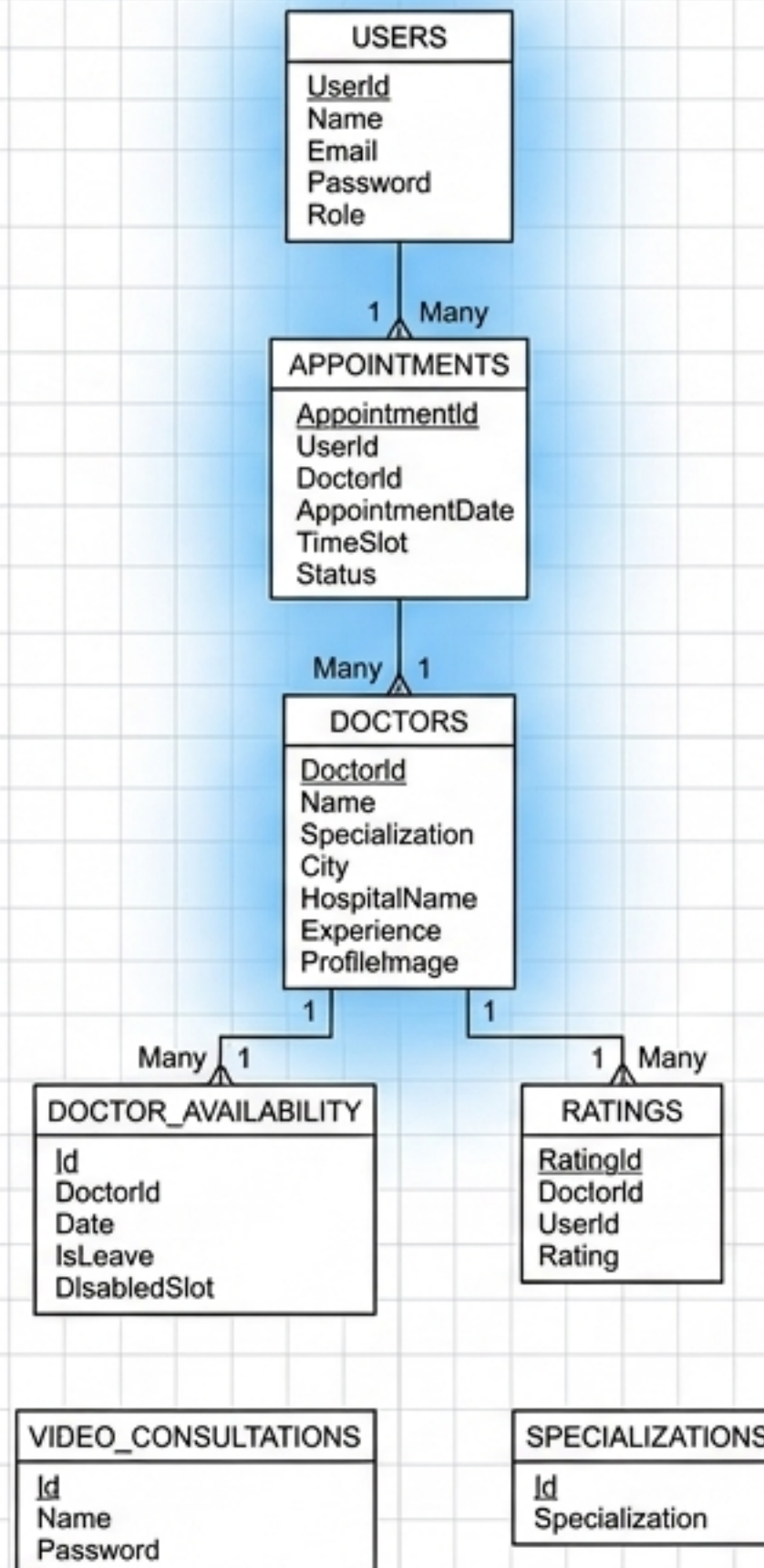
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Confirm: The appointment is permanently saved, and a confirmation is generated.

Organizing and Storing System Data

Relational Integrity

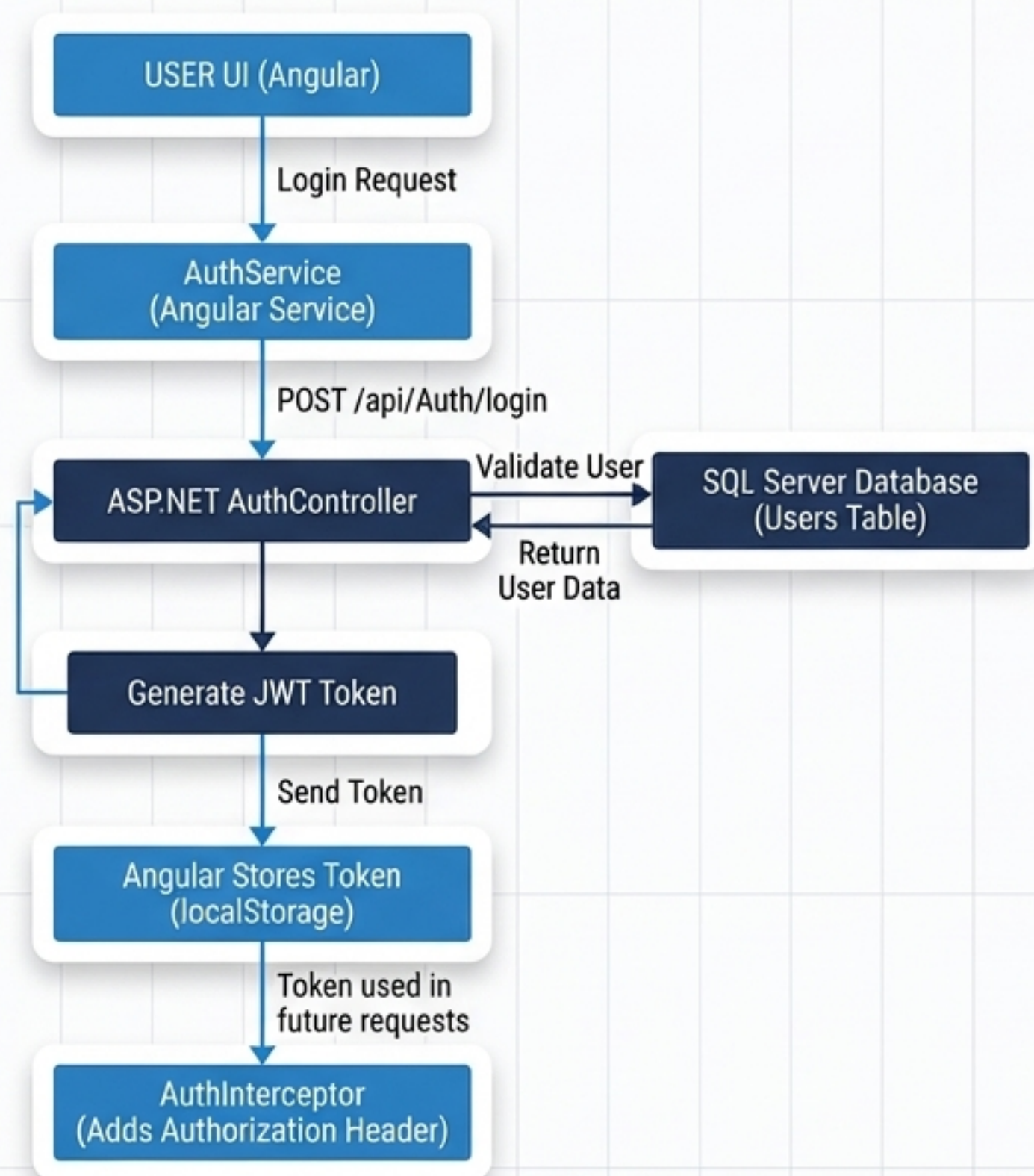
The database is structured to prevent data redundancy and maintain perfect consistency using primary and foreign keys.



The Core Triangle

-  A single User can book multiple appointments over time.
-  A single Doctor manages multiple appointments and receives multiple ratings.
-  The Appointment table acts as the critical bridge linking patients to their chosen healthcare providers.

Keeping User Data Safe and Secure



The JWT Cycle:

- Verify**
1. The ASP.NET AuthController validates user credentials against the SQL database.
- Issue**
2. The server generates a unique JSON Web Token (JWT) containing the user's ID and role.
- Store**
3. Angular securely stores this token locally in the browser.
- Access**
4. An Angular Interceptor automatically attaches this 'digital key' to all future requests, ensuring only authorized users can access sensitive endpoints.

Key Features for Users and Administrators



Patient Features (The Front-of-House)

- ✓ Register and secure login.
- ✓ Search and filter doctors by city, specialty, and rating.
- ✓ Book available time slots and view upcoming appointments.



Admin Features (The Back-of-House)

- ✓ Add, edit, or delete doctor profiles.
- ✓ View all registered system users.
- ✓ Oversee the global appointment roster and manually mark attendance (Present/Absent).

Enhancing the Platform with Real-Time Communication



Video Consultations (Telemedicine):

Introduced remote, real-time video sessions allowing patients and doctors to connect from anywhere, reducing the dependency on physical clinic visits.



SignalR Integration (Live Sync):

Implemented WebSockets to enable bidirectional communication. Appointment status updates (like a doctor marking a patient 'Present') now reflect instantly on the user's screen without requiring a manual page refresh.

Managing Doctor Availability and Profiles

Doctor Schedule

Sun	Mon	Tue	Wed	Thu	Fri	Sat
5	6	7	8	9	10	11
12	Leave					13
18	19	20	21	23	24	25
9:00 AM	∅	11:30 AM	11:30 AM	2:00 PM		
10:30 AM	1:00 AM	∅	2:00 PM	3:00 PM		

Dynamic Calendar Management: Doctors can proactively manage their own schedules by marking leave days and disabling specific time slots, preventing patients from booking invalid appointments.

Doctor Profile Badge



Dr. Alex Chen

Cardiologist

10

Years
Experience

Enhanced Expertise Profiles: Added detailed 'Experience' metrics (e.g., years in practice) to doctor profiles, building patient trust and empowering users to make highly informed healthcare decisions.

Summary of Project Achievements

- ✓ **Full-Stack Mastery:** Successfully architected and deployed a highly scalable, 3-tier healthcare application using Angular, .NET Core, and SQL Server.
- ✓ **Robust Security & Structure:** Implemented airtight JWT authentication and engineered a perfectly normalized relational database using Entity Framework Core.
- ✓ **Real-World Readiness:** Upgraded foundational CRUD operations with advanced, real-world telemedicine capabilities, including SignalR live-syncing and remote video consultations.
- ✓ **Mission Accomplished:** Delivered a seamless digital bridge that successfully eliminates the friction of traditional medical booking for both patients and administrators.